

Curriculum Vitae

Tung Yan “Esther” Liu, Ph.D.

EDUCATION

Texas A&M University , College Station, TX	2023
<i>Ph.D. in Materials Science and Engineering</i> (Advisor: Michael Demkowicz)	
Dissertation: Atomistic Studies of Intergranular Damage and Transport in Metals	
Texas A&M University , College Station, TX	2018
<i>Master of Science in Materials Science and Engineering</i> (Advisor: Michael Demkowicz)	
Thesis: Atomistic Simulations of the Viscoelastic Response of a Model Single-Crystal Equiatomic Solid Solution	
University of Texas at Austin , Austin, TX	2016
<i>Bachelor of Science in Physics (Honor Thesis)</i> (Advisor: Richard Matzner)	
Thesis: “The Estimated Merger Rates of GW150914-like Binary Black Holes from the Astrophysical Argument Perspective and Their Implications”	

SKILLS

Python (NumPy, SciPy, scikit-learn), MATLAB, LAMMPS (molecular dynamics), VASP (DFT), Bash/Linux, SLURM and High Performance Computing, OVITO, HYADES (hydrocode)

PROFESSIONAL EXPERIENCE

Johns Hopkins University , Baltimore, MD	Oct. 2024 – Present
<i>Postdoctoral Fellow, Hopkins Extreme Materials Institute</i>	
- Develop physics-informed Gaussian process regression equation-of-state (EOS) models for materials under extreme pressure and temperature, enabling uncertainty quantification for shock simulations and experimental designs - Support representation learning studies for local atomic environment analysis and machine-learned interatomic potential development for amorphous materials, with a focus on metallic glass	
Texas A&M University , College Station, TX	Aug. 2018 – Oct. 2024
<i>Research Assistant, Department of Materials Science and Engineering</i>	
- Investigated helium bubbles’ evolution and deformation mechanisms in Nickel-based alloy using large-scale molecular dynamics simulations, providing insight into failure of nuclear reactor components - Reduced computational cost by 70× by optimizing simulation workflows in LAMMPS and parallel computing on HPC platforms, enabling efficient study of defect formation energies for studying atom transport at grain boundaries - Conducted parametric studies on loading, temperature, and defect density to analyze sensitivity of system behavior, informing modeling assumptions on material deformation - Collaborated with experimentalists to integrate microscopy data (SEM, TEM, EBSD, EDX) into modeling workflows, improving validation of structure–property relationship in tri-layered refractory metal (Nb, V, Mo)-copper composites	
<i>Master Student Researcher, Department of Materials Science and Engineering</i>	Aug. 2016 – May 2018
- Conducted 2,000+ cyclic loading simulations to characterize viscoelasticity in equi-atomic solid solutions (ESS) - Demonstrated effect of atomic misfit, loading frequency, temperature, and model size in ESS’s mechanical response	

AWARDS AND HONORS

Moore Foundation Postdoctoral Fellow at JHU, The Gordon and Betty Moore Foundation	2026
Finalist, Empower Your Pitch – Postdoctoral session , Johns Hopkins University	2025
1st place, Doctoral Research Impact Pitch Competition , American Society for Composites	2023
Best M.S. Thesis , Department of Materials Science and Engineering, Texas A&M University	2018

PEER-REVIEWED PUBLICATIONS

1. **T. Y. Liu**, M. J. Demkowicz, “Effect of free surfaces on localized plastic deformation in single-crystal nickel containing helium bubbles and radiation-induced self-interstitial atom clusters,” **Journal of Nuclear Materials**, 587 (2023) 154741.
2. **T. Y. Liu**, M. J. Demkowicz, “Effect of grain boundaries and rigid inclusions on plasticity in nickel bicrystals containing helium bubbles and radiation-induced self-interstitial atom clusters,” **Journal of Nuclear Materials**, 594 (2024) 155030.
3. E. Sheu*, **T.Y. Liu***, D. J. Williams, J. K. Baldwin, M. J. Demkowicz, “Permeation of niobium through grain boundaries in copper,” **Acta Materialia**, 274 (2024) 120002. *Co-first author.
4. **T.Y. Liu***, E. Sheu*, D. J. Williams, J. K. Baldwin, M. J. Demkowicz, “Reduction of vanadium diffusivity within copper grain boundaries due to enhanced binding,” **Scripta Materialia**, 258 (2025) 116515.
5. J. C. Mwangeli, **T. Y. Liu**, M. J. Demkowicz, “Temperature-dependence of vacancy/self-interstitial recombination volumes in copper,” **Physical Review Materials** 9, 023601 (2025).
6. **T. Y. Liu**, H. Sharma, S. Ali, M. D. Shields, “Physics-Constrained Gaussian Process Regression for Uncertainty Quantification of Equation-of-State Models,” *in preparation* (2026)

SELECTED PRESENTATIONS

Talks:

1. Uncertainty-Aware, Physics-Constrained Gaussian Process Regression for Equation-of-State Modeling. *MS&T26: Materials Science & Technology*: Pittsburgh PA, October 4 – 7, 2026
2. [Invited talk] Physics-constrained Gaussian Process Regression Equation-of-State Model for Shock Simulations. *American Physical Society SCCM 2025*: Washington, DC, Jun. 22 – 27 2025.
3. Why this way? An atomistic study on diffusion pathway in nanocomposite of immiscible metals. *The 6th High Performance Research Computing (HPRC) Symposium*: College Station, TX, May 18 – 19, 2023.
4. Atomistic simulation of plastic deformation in nickel bi-crystals containing helium bubbles. *The Society of Engineering Science (SES) 2022 Annual Meetings*: College Station, TX, Oct. 15 – 17, 2022.
5. Molecular dynamics simulation of deformation in nickel containing helium bubbles. *10th International Conference on Multiscale Materials Modelling*: Baltimore, MD, Oct. 2 – 7, 2022.
6. Atomistic simulation of plastic deformation in nickel containing helium bubbles. *Mach Conference*: Virtual, Apr. 6 – 8, 2022.
7. Plastic deformation in nickel bicrystals containing helium bubbles. *The Minerals, Metals & Materials Society (TMS) 2022 Annual Meetings & Exhibition*: Anaheim, CA, Feb. 27 – Mar. 3, 2022.
8. Molecular dynamics simulations of the plastic deformation response of Ni single crystals containing He bubbles and radiation-induced self-interstitial atom clusters. *The TMS 2021 Annual Meetings & Exhibition*: Virtual, Mar. 15 – 18, 2021.
9. Modeling slip-induced crack initiation in nickel containing nano-scale helium bubbles. *Canadian Nuclear Laboratories Workshop*: Toronto, Canada, Apr. 9, 2019.

Posters:

1. Physics-Constrained Gaussian Process Regression Equation-of-State Model for Shock Simulations. *Artificial Intelligence for Materials Science (AIMS) Workshop*: National Institute of Standards & Technology (NIST), Gaithersburg, MD, July 9 – 10, 2025.
2. Modeling Slip Bands in Nickel Containing Nano-scale Helium Bubbles. *The TMS 2020 Annual Meetings & Exhibition*: San Diego, CA, Feb. 24 – 28, 2020.